

Differentiation - Set 1 - Solutions

Specification MM6.1-MM6.3, MM8.5.

Answer key

Question	Answer	Topic
1	D	Differentiation
2	C	Differentiation
3	F	Differentiation
4	B	Differentiation
5	F	Differentiation
6	B	Differentiation
7	A	Differentiation
8	E	Differentiation
9	C	Differentiation
10	A	Differentiation
11	D	Differentiation
12	B	Differentiation

Worked solutions

Question 1 DIF-01 • Differentiation • spec MM6.2

Find $\frac{dy}{dx}$ when $y = \frac{(3x-2)^2}{x^2}$.

Answer: D

$$y = 9 - 12x^{-1} + 4x^{-2}, \text{ so } y' = 12x^{-2} - 8x^{-3}.$$

Question 2 DIF-02 • Differentiation • spec MM6.3

Find the coordinates of the stationary points of $y = x^3 - 3x^2 - 9x + 5$ and state their nature.

Answer: C

$$y' = 3x^2 - 6x - 9 = 3(x-3)(x+1). \quad y'' = 6x - 6: \text{ negative at } x = -1 \text{ (max, } y = -1 - 3 + 9 + 5 = 10\text{), positive at } x = 3 \text{ (min, } y = 27 - 27 - 27 + 5 = -22\text{)}.$$

Question 3 DIF-03 • Differentiation • spec MM6.3

The tangent to $y = x^2 - 4x + 1$ at the point where $x = 3$ meets the y-axis at

Answer: F

$$y' = 2x - 4 = 2 \text{ at } x = 3; \text{ the point is } (3, -2). \text{ Tangent: } y + 2 = 2(x - 3), \text{ so at } x = 0, y = -8.$$

Question 4 DIF-04 • Differentiation • spec MM6.3

Find the equation of the normal to the curve $y = \sqrt{x}$ at the point $(4, 2)$.

Answer: B

$$y' = \frac{1}{2\sqrt{x}} = \frac{1}{4} \text{ at } x = 4, \text{ so the normal has gradient } -4: y - 2 = -4(x - 4), \text{ i.e. } y = -4x + 18.$$

Question 5 DIF-05 • Differentiation • spec MM6.3

For what range of values of x is $f(x) = 2x^3 - 15x^2 + 24x + 1$ decreasing?

Answer: F

$$f'(x) = 6x^2 - 30x + 24 = 6(x-1)(x-4) < 0 \text{ for } 1 < x < 4.$$

Question 6 DIF-06 • Differentiation • spec MM6.1

The displacement s metres of a particle at time t seconds is $s = t^3 - 6t^2 + 9t$. At what times is the particle momentarily at rest?

Answer: B

$$\text{Velocity} = \frac{ds}{dt} = 3t^2 - 12t + 9 = 3(t-1)(t-3) = 0 \text{ at } t = 1 \text{ and } t = 3.$$

Question 7 DIF-07 • Differentiation • spec MM6.3

The curve $y = ax^2 + bx$ has a stationary point at $(2, -8)$. Find $a + b$.

Answer: A

$$y' = 2ax + b = 0 \text{ at } x = 2: b = -4a. \quad y(2) = 4a + 2b = 4a - 8a = -4a = -8 \Rightarrow a = 2, b = -8. \quad a + b = -6.$$

Question 8 DIF-09 · Differentiation · spec MM6.3

A rectangle has perimeter 40. What is its maximum possible area?

Answer: E

Sides x and $20 - x$: $A = 20x - x^2$, $A' = 20 - 2x = 0$ at $x = 10$, giving $A = 100$ (a square).

Question 9 DIF-10 · Differentiation · spec MM6.3

The curve $y = x^3 + px^2 + qx$ has stationary points at $x = 1$ and $x = 3$. Find q .

Answer: C

$y' = 3x^2 + 2px + q = 3(x - 1)(x - 3) = 3x^2 - 12x + 9$. So $q = 9$ (and $p = -6$).

Question 10 DIF-12 · Differentiation · spec MM6.3

The line $y = 4x + c$ is a tangent to $y = x^2 + 2x + 5$. Find c .

Answer: A

$2x + 2 = 4 \Rightarrow x = 1$, $y = 8$. $8 = 4 + c \Rightarrow c = 4$.

Question 11 DIF-14 · Differentiation · spec MM6.2

Find $\frac{d^2y}{dx^2}$ when $y = \frac{1}{x^2} + 4\sqrt{x}$.

Answer: D

$y' = -2x^{-3} + 2x^{-1/2}$; $y'' = 6x^{-4} - x^{-3/2}$.

Question 12 DIF-17 · Differentiation · spec MM6.3

The curve $y = x^2 + kx$ (with $k < 0$) crosses the x -axis at the origin and at one other point P . The gradient of the curve at P is 6. Find k .

Answer: B

$P = (-k, 0)$. $y' = 2x + k = -2k + k = -k = 6$, so $k = -6$.